Real-World Evidence Against Ground Truth

Validating a Causal Inference Pipeline on Semi-Synthetic EHR Data

Dr. Valerio Ganci
vganci26@gmail.com

ABSTRACT

Propensity-score methods are the workhorse of real-world evidence studies, yet their validity is normally unverifiable: observational data never reveal the true causal effect an estimator is meant to recover. This paper sidesteps that problem by construction. Baseline covariates for 1,551 type 2 diabetes patients on metformin come from Synthea-generated synthetic EHR data; treatment assignment (SGLT2 inhibitor vs. DPP-4 inhibitor) and outcome (heart failure hospitalization) are then semi-synthetically injected via a logistic model and an exponential proportional-hazards model, with a hazard ratio (HR) of 0.67 and realistic confounding.

Because the ground truth is known, the pipeline can be scored directly. A naive unadjusted Cox model, biased by confounding, recovers an HR of 0.481 (95% CI 0.349–0.663); a stabilized Inverse Probability of Treatment Weighting (IPTW) model and a doubly-robust estimator recover 0.496 (0.351–0.702) and 0.482 (0.329–0.706), both consistent with the truth. A 500-seed Monte Carlo replication confirms this: IPTW achieves 94.6% empirical CI coverage against a 95% target, versus 70.6% for the naive estimator, while doubly-robust proves unstable, failing to fit in 11.6% of seeds and, in a further 3.6%, converging to an extreme, unusable hazard ratio when a near-zero propensity score inflates the IPTW weights.

A negative control outcome and an IPTW E-value of 2.20 further support the adjusted estimates. A leave-one-out sweep over the six confounders shows omission severity is not simply predicted by a covariate's own effect size: dropping age hurts IPTW's calibration most, by a wide margin, since it has no strongly correlated proxy among the other covariates; dropping creatinine, BMI, or eGFR forms a moderately-damaging middle tier whose internal order inverts coefficient size, since eGFR and creatinine ($r = -0.546$) partly proxy for each other while BMI, lacking a comparably strong proxy, is more damaging to drop despite a smaller coefficient than eGFR's; and dropping HbA1c or systolic blood pressure, the two smallest-effect covariates, is comparatively harmless outright. Together, these results show a standard causal pipeline can recover a known effect from confounded data while exposing failure modes, doubly-robust instability chief among them, invisible to a single real-data analysis.

KEYWORDS

Real-world evidence; Target trial emulation; Propensity scores; Inverse probability of treatment weighting; Doubly-robust estimation; Synthetic EHR data

1 INTRODUCTION

Real-world evidence (RWE) studies routinely ask whether a drug already in clinical use is safer or more effective than an existing alternative, using data that were never collected with this question in mind: electronic health records (EHRs), insurance claims, or registries [19]. Propensity-score methods, most commonly inverse probability of treatment weighting (IPTW) and doubly-robust estimation, are the standard tools for extracting a causal answer from such confounded, non-randomized data [3]. They are used to inform prescribing guidance, payer decisions, and post-marketing safety assessments, so a wrong estimate is not a minor inconvenience: it can misdirect clinical practice at scale [19].

The trouble is that these methods are almost never validated against a known answer. In a real observational dataset, the true causal effect is unobservable; an analyst can check that covariates are balanced after weighting, run a negative control, or compute a sensitivity bound, but cannot directly confirm that the estimator recovered the right number, because there is no ground truth to recover. This is a well-known limitation of the field, and it makes methodological claims ("IPTW correctly adjusts for confounding under X assumptions") difficult to demonstrate empirically rather than just argue analytically [11].

This project builds an end-to-end pipeline (available at here) that closes this gap by construction. It runs a target trial emulation, comparing SGLT2 inhibitors to DPP-4 inhibitors on heart failure hospitalization risk in patients with type 2 diabetes, on a cohort whose baseline covariates come from Synthea-generated synthetic EHRs, but whose treatment assignment and outcome are semi-synthetically injected on top of those covariates by the investigator, using a known confounding structure and a known treatment effect. Because the true hazard ratio is fixed in advance, the pipeline's estimators can be scored directly against it, something no purely observational analysis can do.

The specific clinical scenario is not incidental to the validation exercise; it was chosen because it is close to the best-case setting a propensity-score pipeline could hope to be tested in, which is precisely what a methods-validation study needs. Three choices matter here. First, the population: type 2 diabetes is one of the most densely instrumented chronic conditions in any EHR or claims system, with frequent encounters, routine laboratory monitoring (HbA1c, renal function, lipids), and long, continuous treatment histories, as reflected in consensus clinical practice guidelines that mandate regular structured monitoring and long-term pharmacologic management [5]. A realistic new-user cohort with a rich baseline covariate set therefore reflects data that plausibly exists, rather than requiring invented variables. Second, the comparator: SGLT2 inhibitors and DPP-4 inhibitors are both standard second-line add-on therapies to metformin, prescribed to overlapping patient populations for the same glycemic indication [5]. This is exactly the kind of

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Dr. Valerio Ganci.

clinical uncertainty that produces confounding by indication rather than confounding by contraindication [17, 18]: the two treatment groups can plausibly overlap on their propensity scores, satisfying positivity [23], rather than being separated by an obvious clinical rule (such as a strict renal-function cutoff) that would make any statistical adjustment ineffective. Third, the outcome: heart failure hospitalization is the single most consistently replicated benefit of SGLT2 inhibitors across independent cardiovascular outcomes trials and real-world studies [16, 24, 25], which means the injected effect size used in this pipeline is not an arbitrary number chosen for convenience, but a literature-grounded hazard ratio that a real analyst could recognize and sanity-check against. Together, these choices give the simulation a cover story that is clinically coherent rather than a purely mechanical exercise, while keeping the actual object of study exactly where it belongs: the behavior of the estimators, not a claim about diabetes care.

Concretely, the pipeline answers three nested questions:

(1) Recovery: When the confounding structure and true effect are known by construction, do standard propensity-score estimators (naive, IPTW, doubly-robust) recover the injected hazard ratio, and how large is the bias left uncorrected by each?
(2) Calibration: Beyond a single realization, do the confidence intervals produced by each estimator achieve their nominal coverage rate across many independent draws from the same data-generating process, and are there failure modes (non-convergence, unstable weights) that a single run would not reveal?
(3) Robustness: How does estimator performance degrade under conditions that mimic real-world imperfection, specifically a propensity model that omits a true confounder, and how do auxiliary diagnostics (negative control outcome, E-value) support or qualify the main result?

The end deliverable is not a single hazard ratio but a full validation exercise, quantifying exactly how well, and under what conditions, a standard causal inference pipeline can recover an effect it was built to find.

2 DATA

The pipeline is built on a single public data source, Synthea, plus a layer of investigator-injected data generated on top of it. This section describes the source, why it was chosen, what it contributes, and how the semi-synthetic layer is constructed conceptually (the injection mechanics themselves are covered in Section 3).

2.1 Synthea Synthetic EHR Data

Synthea is an open-source synthetic patient generator that produces realistic, but not real, EHR data: demographics, conditions, medications, encounters, and laboratory observations, generated from statistically modeled disease and care-pathway modules rather than sampled from actual patients [12]. Because no real patient data is involved, it can be used, shared, and published without any privacy or consent constraint, while still exhibiting much of the structure (coding systems, visit cadence, comorbidity co-occurrence, missingness patterns) that makes real EHR data hard to work with.

This project uses a Synthea v3.3.0 population of Illinois patients ($n \approx$ 115,586), generated with a fixed random seed for reproducibility. The following raw tables are consumed:

- Patients table: one row per synthetic patient, providing demographics used as baseline covariates.
- Conditions table: one row per diagnosed condition per patient, used to identify the type 2 diabetes mellitus (T2DM) cohort, prior heart failure, and diabetes-related complications.
- Medications table: one row per dispensed medication per patient, used to identify metformin new-users (the active-comparator anchor) and to confirm the near-absence of SGLT2 inhibitor and DPP-4 inhibitor prescribing that motivates the injection design described below.
- Observations table: one row per recorded laboratory or vital-sign measurement, used to extract five baseline clinical covariates (BMI, HbA1c, systolic blood pressure, estimated glomerular filtration rate (eGFR), and creatinine) in a window around each patient's cohort-entry date; together with age, these form the six baseline covariates that enter the treatment-assignment and outcome models (Section 3.2).

All cohort-defining and covariate-defining identifiers are code-based (SNOMED CT for conditions, RxNorm for medications, LOINC for observations) rather than free-text matches, so that cohort construction is deterministic and auditable; the exact codes used are enumerated in Section 3.

2.2 The Semi-Synthetic Injection Layer

Synthea's medication-prescribing modules do not meaningfully differentiate between SGLT2 inhibitors and DPP-4 inhibitors, so treatment assignment cannot simply be read off the raw data the way it would be in a real claims dataset. Rather than treat this as a limitation, the pipeline turns it into the project's central design choice: Synthea baseline covariates are kept as-is, while treatment assignment (SGLT2 inhibitor vs. DPP-4 inhibitor) and the outcome (heart failure hospitalization) are generated by the investigator, under a fully specified and disclosed data-generating process, with a known confounding structure and a known true hazard ratio. This makes the resulting dataset semi-synthetic in a specific sense that is central to the analysis: covariates and their joint distribution are as real as Synthea produces, while treatment and outcome are simulated on top of them, which is what allows Section 3 and Section 5 to compare each estimator's output against a known ground truth, something not possible when treatment and outcome are both observed rather than assigned by the investigator.

3 METHODOLOGY

This section explains, stage by stage, how raw Synthea data is transformed into a semi-synthetic, ground-truth-validated causal analysis, and, for every major step, why that specific method was chosen. The methodology is split into five subsections, mirroring the pipeline's five run stages:

3.1 Cohort construction

3.2 Ground-truth injection

3.3 Causal estimation

3.4 Monte Carlo calibration validation

3.5 Misspecification sensitivity

3.1 Cohort Construction

Every cohort-defining decision below was preceded by a dedicated exploratory data analysis (EDA) step, so that inclusion/exclusion logic is justified by an observed property of the data rather than assumed.

3.1.1 Type 2 diabetes identification. Patients are identified as type 2 diabetes mellitus (T2DM) cases using SNOMED CT condition codes rather than free-text description matching. EDA showed that description-based substring matching misses the base diagnosis due to word-order variation ("Diabetes mellitus type 2" vs. "type 2 diabetes") and Roman-numeral variants ("type II"), whereas every code maps 1:1 to exactly one description in this dataset, making code-based matching immune to phrasing. A patient is included if they have the base T2DM diagnosis code (44054006) or any of eight T2DM-specific complication codes (listed here); EDA showed that 73.9% of patients with a complication code never had a base-diagnosis row on file, so requiring the base diagnosis alone would wrongly exclude the majority of true T2DM patients. This step yields 20,962 T2DM patients.

Four adjacent codes were deliberately excluded from this inclusion rule rather than silently ignored: prediabetes (code 15777000, 42,540 patients), a distinct, earlier-stage condition than T2DM itself [2]; a kidney-disorder-due-to-diabetes (code 127013003, 17,754 patients) that does not specify diabetes type and would risk including type 1 diabetes patients; a cystic-fibrosis-related diabetes (code 427089005, 23 patients), a mechanistically distinct disease entity despite the shared name [14]; and a diabetes-screening encounter (code Z13.1, 15 patients), which records a screening visit rather than a diagnosis [21].

3.1.2 Metformin new-user, active-comparator cohort. The cohort is anchored on metformin as a new-user, active-comparator design: rather than comparing SGLT2 inhibitor users to non-users, which conflates the treatment effect with the confounding structure of who gets treated at all, the comparison is restricted to metformin new-users, among whom the later choice of a second-line agent (SGLT2i vs. DPP-4i) is the exposure of interest. Two independent passes over the medication data, a broad, unbiased ranking of every medication prescribed to T2DM-cohort patients by unique-patient count, and a targeted regex search for antidiabetic drug-name patterns (listed here), agree exactly on six antidiabetic drugs (codes: 106892, 860975, 311034, 897122, 865098, 1373463), confirming the six-drug list is exhaustive rather than an artifact of an incomplete search. This cross-check matters because the broad ranking alone would have been misleading: metformin (3,995 patients) narrowly falls just outside that ranking's top 15 medications by volume, and canagliflozin, the sole SGLT2 inhibitor present at all, appears in only 6 patients, orders of magnitude below the ranking's visible cutoff. Volume is concentrated in metformin and an insulin 70/30 mix (10,527 patients); the remaining four drugs, including the two active-comparator classes of ultimate interest, are rare in raw form, confirming that the exposure of interest must itself be injected rather than observed (Section 2). Metformin new-user status is enforced by requiring metformin to be each patient's first antidiabetic drug; only 16 of 3,995 metformin-exposed patients (0.4%) had a different antidiabetic drug first, so this restriction excludes very few patients while enforcing a clean new-user design. This step yields 3,979 metformin new-users.

3.1.3 Excluding prior heart failure. Patients with a pre-existing heart failure diagnosis (SNOMED codes for "chronic congestive heart failure" and "heart failure", treated as equally disqualifying) diagnosed on or before their metformin index date are excluded, since such patients cannot experience the incident heart failure hospitalization outcome as a genuinely new event. A preliminary EDA check found that, dataset-wide, 2,919 unique patients carry a heart failure diagnosis under either code, of whom 225 are also present in the 3,979-patient metformin cohort at some point in their record (i.e. before, at, or after their metformin index date); applying the actual on-or-before-index timing restriction narrows this down to the 15 patients this exclusion step removes, since most of the 225 developed heart failure only after entering the cohort and therefore remain eligible to contribute an incident heart failure event later. This step yields 3,964 patients.

3.1.4 Baseline covariate extraction. Five baseline clinical covariates are extracted from the Observations table in a 40-day window after each patient's metformin start date (index date): body mass index (BMI), hemoglobin A1c (HbA1c), systolic blood pressure, estimated glomerular filtration rate (eGFR), and creatinine; for each patient the most recent value not later than 40 days after the index date is kept. Most labs are recorded shortly after, rather than before, metformin start, a timing artifact of Synthea's simulated encounter ordering rather than missing data.

The 40-day cutoff itself is not an arbitrary round number: the percentile breakdown of each covariate's nearest-observation distance from the index date, computed across all 3,964 no-prior-heart-failure cohort patients, shows the same pattern for every covariate. Roughly 40–43% of patients have their nearest observation within 28 days of the index date, after which the distribution jumps directly to a second cluster starting around day 147–154 and plateauing near day 371, with essentially no density in between. The window is set just above the first cluster's upper edge (day 28) and well below the second cluster's onset (day 147), so it captures the same-visit-cluster labs and excludes the later ones, rather than landing at an arbitrary point on a continuous curve. This is also why the window captures only 40–43% of the no-prior-heart-failure cohort per covariate (1,676–1,694 of 3,964 patients, depending on the covariate) rather than a higher fraction: the later cluster of labs is deliberately excluded by design, since a lab drawn 5–12 months after starting metformin is not a credible baseline measurement, even though including it would have reduced the complete-case cohort's missingness.

Three data-quality corrections, all justified by EDA, are applied before the covariates are used:

(1) Canonical variant filtering: eGFR and creatinine each have multiple description/units variants recorded under the same LOINC code, with materially different value distributions between variants. Since the majority variant alone already covers 100% of the cohort for both labs, only that variant is kept for each and the minority variants are discarded rather than reconciled.

(2) Scale correction: across all eGFR variants, EDA found 655 records tightly clustered at implausible values of 1.0–1.9, all recoverable by multiplying by 100. This was confirmed, via LOINC/FHIR documentation, to have no legitimate alternate scale or unit convention for this code, so these values are corrected by a factor of 100 rather than discarded; after canonical-variant filtering (above) removes the non-canonical records first, 654 of these fall within the canonical variant and are the ones actually rescaled.

(3) Duplicate-timestamp resolution: for a subset of patients, the raw Observations table contains multiple eGFR or creatinine records at the exact same timestamp, same encounter, same LOINC code, description, and units, but with different recorded values, a duplication confirmed present in Synthea's raw output itself (dataset-wide, 155,470 such same-timestamp, conflicting-value record groups) rather than an artifact of this pipeline's loading or filtering logic. Since these duplicates can tie exactly on distance from the index date, selecting a single "closest to index" value among them non-deterministically (via an unstable sort) or by an arbitrary rule (e.g. always the minimum) both proved unsatisfactory: the former made the extracted covariates depend on incidental row order, and the latter introduced a systematic directional bias in the extracted values. The pipeline instead averages same-timestamp duplicate values for a given patient and covariate before selecting the closest-to-index observation, which is deterministic and does not favor either duplicate over the other.

After correction, remaining values outside a clinically plausible range per covariate are set to missing rather than silently retained. These ranges (Table 1) are deliberately generous clinical bounds [7] meant to catch obvious data errors, such as the eGFR unit-scaling artifact above or an implausible outlier from a transcription error, rather than tight, disease-specific reference ranges that would flag legitimate but unusual clinical values as missing.
The analysis proceeds on the complete-case subset of patients with all five clinical covariates present, yielding the final analytic cohort of 1,551 patients. The cohort attrition funnel is summarized in Table 2.

Table 1: Ranges used for implausible-value exclusion.

Covariate	Range
BMI (kg/m^2)	10 – 80
HbA1c (%)	3 – 20
Systolic BP (mmHg)	60 – 250
eGFR (mL/min/1.73m^2)	2 – 200
Creatinine (mg/dL)	0.1 – 15

Table 2: Cohort attrition funnel

Stage	Patients
T2DM patients	20,962
Metformin new-users	3,979
No prior heart failure	3,964
Complete-case	1,551

Table 3 reports descriptive statistics for the final complete-case analytic cohort. These figures serve two purposes: they are a plausibility check on the cohort produced by the preceding filters (e.g. a mean HbA1c of 7.3% and mean BMI of 30.7 kg/m^2 are consistent with a real-world T2DM population starting metformin [9]), and they are the empirical basis on which the injection stage's covariate standardization (Section 3.2) is performed, since covariates are z-scored using this cohort's own mean and standard deviation rather than an external reference population.

Table 3: Final cohort baseline characteristics

Covariate	Mean	SD	Median	Range
Age (years)	52.0	12.1	51.0	18.1 – 97.4
BMI (kg/m^2)	30.7	2.0	30.3	27.1 – 53.3
HbA1c (%)	7.3	1.3	7.6	3.0 – 12.0
Systolic BP (mmHg)	124.6	17.5	124.0	67.0 – 175.0
eGFR (mL/min/1.73m^2)	107.1	33.1	110.0	13.7 – 160.4
Creatinine (mg/dL)	1.3	1.0	1.1	0.4 – 14.2

Because the same six covariates enter both the treatment-assignment and outcome models, how they relate to *each other* matters as much as their individual distributions: a covariate that is dropped from the estimated propensity model may still be partly recoverable through a correlated covariate that remains, which changes how much bias its omission actually introduces. Table 4 reports the full pairwise Pearson correlation matrix among the six covariates, computed on the analytic cohort. This matrix is used later, in Section 3.5, to explain why omitting different confounders from the estimated propensity model does not degrade estimation equally.

3.2 Ground-Truth Injection

With a realistic baseline-covariate cohort in hand, treatment assignment and outcome are generated by the investigator under a fully disclosed data-generating process, so that the true causal effect is known and can later be checked against each estimator's output.

Table 4: Covariates pairwise Pearson correlations

Covariate pair	Pearson r
eGFR – creatinine	−0.546
HbA1c – eGFR	0.439
HbA1c – creatinine	−0.418
Age – HbA1c	−0.352
BMI – HbA1c	0.324
Age – eGFR	−0.239
Age – systolic BP	−0.214
HbA1c – systolic BP	0.203
BMI – creatinine	0.132
eGFR – systolic BP	0.116
Creatinine – systolic BP	−0.071
Age – creatinine	0.067
Age – BMI	−0.064
BMI – systolic BP	0.051
BMI – eGFR	−0.023

3.2.1 Treatment assignment. Each patient's probability of receiving an SGLT2 inhibitor (vs. a DPP-4 inhibitor) is modeled with a logistic function of the six standardized (z-scored) baseline covariates:

$$\operatorname{logit} P(\mathrm{SGLT2i}_i) = \beta_0 + \sum_k \beta_k \, z_{ik}$$

where $P(\mathrm{SGLT2i}_i)$ is patient i's probability of being assigned SGLT2i (vs. DPP-4i); β_0 is the intercept, calibrated to hit the target overall uptake rate; k indexes the six baseline covariates (age, BMI, eGFR, HbA1c, systolic BP, creatinine); z_{ik} is patient i's standardized (z-scored) value of covariate k; and β_k is that covariate's fixed assignment coefficient (Table 5). Treatment is then a genuine stochastic draw rather than a deterministic threshold: for each patient, a uniform random number $u_i \sim \mathrm{Uniform}(0, 1)$ is drawn from a seeded generator, and the patient is assigned SGLT2i if $u_i < P(\mathrm{SGLT2i}_i)$ and DPP-4i otherwise, so that two patients with identical covariates and identical assigned probability can still receive different treatments, matching how treatment choice behaves in reality.

Table 5 reports the exact coefficients used, on standardized covariates, for both the treatment-assignment model and the outcome model (Section 3.2.2). Coefficients are chosen to reflect plausible clinical prescribing behavior and disease risk rather than fit to data: for treatment assignment, age is negatively associated with SGLT2i use (older patients more often given a DPP-4 inhibitor) [10], BMI and eGFR are positively associated (SGLT2 inhibitors carry a weight-loss indication and are preferentially avoided in poor renal function) [10], creatinine is negatively associated (consistent with the eGFR direction) [10], systolic blood pressure is weakly positively associated (a modest cardiometabolic-risk signal) [5], and HbA1c is left near zero, since it does not strongly differentiate the choice between these two drug classes in practice [5]. The intercept β_0 is calibrated via bisection search to target an overall SGLT2i uptake of approximately 45% at the cohort's observed covariate distribution. Every covariate that drives treatment assignment also drives the outcome (Section 3.2.2), so all confounding in the simulated data is by design and non-trivial rather than assumed away.

Table 5: Injection-model coefficients

Covariate	Assignment β_k	Outcome β_k
Age	−0.35	0.30
BMI	0.30	0.15
eGFR	0.40	−0.35
HbA1c	0.05	0.20
Systolic BP	0.10	0.15
Creatinine	−0.25	0.25

3.2.2 Outcome simulation. The outcome, incident heart failure hospitalization, is simulated under an exponential proportional-hazards model:

$$h_i(t) = \lambda_0 \cdot \exp\left(\beta_{\mathrm{treat}} \cdot \mathbb{1}(\mathrm{SGLT2i}_i) + \sum_k \beta_k \, z_{ik}\right)$$

where $h_i(t)$ is patient i's hazard of HF hospitalization at time t; λ_0 is the baseline hazard, calibrated to hit the target two-year cumulative incidence; $\mathbb{1}(\mathrm{SGLT2i}_i)$ is an indicator equal to 1 if patient i was assigned SGLT2i and 0 otherwise; β_{treat} is the true log-hazard-ratio for treatment; and, as in the assignment model above, k indexes the six baseline covariates, z_{ik} is patient i's standardized value of covariate k, and β_k is that covariate's fixed outcome coefficient (Table 5). The true treatment effect β_{treat} is fixed to a hazard ratio of 0.67, the midpoint of the range reported across three published cardiovascular-outcomes trials of SGLT2 inhibitors (EMPA-REG OUTCOME, DECLARE-TIMI 58, CVD-REAL, each in the 0.65–0.70 range), so that the injected ground truth is not an arbitrary number but one anchored to real clinical evidence, while remaining explicitly a simulated rather than an estimated effect. Covariate coefficients (Table 5) are set so that age, HbA1c, systolic blood pressure, and creatinine increase heart failure risk, eGFR is protective, and BMI has a modest positive association, matching the same six covariates used in treatment assignment so that all confounding in the simulated data is by the six measured baseline covariates and nothing else. The baseline hazard λ_0 is calibrated, again via bisection search, to target an overall two-year cumulative heart failure hospitalization incidence of approximately 10% [15]. Event times are drawn by inverse cumulative distribution function sampling from each patient's individual exponential hazard, using a random-number generator seeded independently from the treatment-assignment draw (offset by one from the base seed), so that the two stochastic mechanisms are not spuriously correlated with each other through shared randomness; administrative censoring is then applied at a fixed two-year follow-up horizon.

3.2.3 Confounding check. Before proceeding to estimation, a naive (unadjusted) Cox model is fit on the injected data purely as a diagnostic, to confirm that the constructed confounding structure actually biases a naive comparison, rather than assuming it does. Standardized mean difference (SMD) covariate balance between treatment arms is also computed at this stage, prior to any weighting, to quantify the magnitude of imbalance that later needs correcting.

3.3 Causal Estimation

Given the injected dataset, the pipeline runs three treatment-effect estimators and two auxiliary diagnostics, described below.

3.3.1 Propensity scores and estimators. Propensity scores are estimated via logistic regression of treatment on the six baseline covariates. Three hazard ratio estimates are then produced:

(1) Naive: an unadjusted Cox proportional-hazards model of the outcome on treatment alone, included as a baseline to quantify uncorrected confounding bias.
(2) Stabilized IPTW: a Cox model of the outcome on treatment, weighted by stabilized inverse probability of treatment weights, with the marginal treatment prevalence in the numerator, which controls variance relative to unstabilized weights while remaining a consistent estimator of the marginal treatment effect. Weights are not truncated, so the full effect of any extreme propensity scores is retained and visible in downstream diagnostics rather than masked.
(3) Doubly-robust: a Cox model of the outcome on treatment plus all six baseline covariates, additionally weighted by the same stabilized IPTW weights, so that the estimate remains consistent if either the propensity model or the outcome model is correctly specified, but not necessarily both.

The two weighted models (IPTW-weighted and doubly-robust) use robust (sandwich) standard errors to account for the weighting; the naive model, being unweighted, uses the standard Cox partial-likelihood variance.

3.3.2 Covariate balance and the Love plot. Standardized mean differences between treatment arms are recomputed after IPTW weighting for all six covariates and compared against the pre-weighting SMDs from Section 3.2 in a Love plots. A common threshold of $|\text{SMD}| < 0.1$ is used to judge adequate post-weighting balance.

3.3.3 Negative control outcome. As a falsification test, a second outcome is simulated with the same confounding structure and covariate coefficients as the primary outcome, but with the true treatment effect fixed at a hazard ratio of 1.0 (no true effect). Naive and IPTW estimators are re-run on this negative control outcome; a well-behaved analysis should produce confidence intervals that contain the null, since any adjustment method that generates a spurious effect on a variable with no true causal relationship to treatment cannot be trusted on the primary outcome either.

3.3.4 E-value. As a bias-sensitivity diagnostic, a VanderWeele and Ding E-value [22] is computed from the IPTW hazard ratio and from the confidence-interval bound closest to the null, using the standard formula $\text{E-value} = \text{RR} + \sqrt{\text{RR}(\text{RR}-1)}$ where RR is the risk-ratio approximation of the hazard ratio (the hazard ratio itself, or its reciprocal $1/\text{HR}$ when $\text{HR} < 1$, so that $\text{RR} \geq 1$), and E-value is the resulting minimum strength of association, on the risk-ratio scale, that an unmeasured confounder would need with both treatment and outcome, above and beyond the six measured covariates, to fully explain away the observed association.

3.4 Monte Carlo Calibration Validation

A single realization of the injection-and-estimation pipeline can be favorable or unfavorable by chance. To assess whether each estimator is calibrated, rather than merely unbiased in one run, the full injection and analysis pipeline (Sections 3.2–3.3) is repeated end-to-end across 500 independent random seeds, reusing the same fixed analytic cohort of 1,551 patients (baseline covariates are held fixed; only the injected treatment and outcome are redrawn per seed).

For each estimator, two quantities are computed across the 500 seeds: the mean hazard ratio, to assess average bias against the known true value, and the empirical coverage rate, the fraction of seeds whose 95% confidence interval actually contains the true hazard ratio, which should be close to 95% for a correctly calibrated estimator. A mean Love plot, averaging covariate balance across all 500 seeds, is also produced, to distinguish a systematic balance problem from a single unlucky draw.

Seeds are additionally screened for pathological doubly-robust results, flagged when the doubly-robust hazard ratio for that seed falls outside a wide $[0.05, 5.0]$ bound. Flagged seeds are investigated with two targeted diagnostics: a check for quasi-separation among covariates within the event cases (a small-sample condition under which a Cox model's coefficients and standard errors become numerically unstable), and a check for unstable coefficient standard errors (flagged when any covariate's standard error exceeds 2), together with the seed's minimum propensity score and maximum IPTW weight, since a single near-zero propensity score is often the direct cause of both an extreme weight and an unstable downstream model fit.

3.5 Misspecification sensitivity

Every result described so far uses a correctly specified propensity model: the model estimated from data includes exactly the same six covariates used to generate the true confounding structure. Real-world propensity models are never guaranteed to be correctly specified, since an analyst can only include confounders they know to measure. To quantify how the pipeline degrades under this realistic condition, the full 500-seed Monte Carlo procedure (Section 3.4) is re-run six times, once per covariate, each time with exactly one true confounder deliberately omitted from the *estimated* propensity model while the injected ground truth continues to depend on all six covariates as in the baseline case. The test will directly assess IPTW's best-known vulnerability, that its consistency depends on including all true confounders in the propensity model, letting the resulting bias and coverage degradation be measured, compared across covariates, and explained, rather than merely asserted for one or two illustrative cases.

4 LIMITATIONS

A pipeline that only shows its strengths is less credible than one that also states clearly where it approximates, where it assumes, and where it is simply out of scope. In this Section, the limitations of the work are presented, grouped into three categories: input data, methodology, and the results themselves.

4.1 Input data limitations

4.1.1 Synthea is synthetic, not real, patient data. Every baseline covariate value ultimately comes from Synthea's generative modules, not from an observed patient. Synthea is built to reproduce realistic marginal distributions and plausible disease/care-pathway trajectories, but it does not reproduce every subtle statistical dependency present in real EHR data, and its modules can carry their own generation artifacts. One such artifact drives the baseline window itself: EDA showed that most lab values are recorded shortly *after*, rather than before, the metformin index date, a consequence of Synthea's simulated encounter ordering rather than a property that would necessarily hold in real claims or EHR data.

4.1.2 Complete-case cohort carries substantial missingness. The final analytic cohort (1,551 of 3,964 eligible patients, roughly 60% missingness) is restricted to patients with all five clinical baseline covariates recorded in the 40-day window. This missingness is itself a Synthea lab-visit-cadence artifact rather than random measurement dropout, but the complete-case approach used here does not model or adjust for it, so the analytic cohort may not be perfectly representative of the full eligible population, and a comparable real-world study would need to explicitly justify (or relax) a complete-case assumption.

4.1.3 SGLT2 and DPP-4 inhibitor prescribing is not observed. Synthea's medication modules do not meaningfully differentiate between these two drug classes, which is precisely why treatment assignment is injected rather than drawn from the data (Section 2). This means the pipeline's realism rests entirely on the plausibility of the investigator-specified injection coefficients and calibration targets (Section 3.2), not on any observed prescribing pattern; a materially different, but equally defensible, choice of coefficients would change the confounding strength and event rate without the pipeline itself flagging that choice as wrong.

4.2 Methodology limitations

4.2.1 Injection coefficients are investigator-specified, not externally fit. The logistic treatment-assignment coefficients and the exponential outcome-model coefficients (Section 3.2) were chosen by the investigator to be clinically plausible and, for the treatment hazard ratio, anchored to published cardiovascular-outcomes trials, but they were not fit to any external dataset of actual SGLT2i/DPP-4i prescribing or outcomes. This is an intentional feature of a ground-truth validation study, since the true effect must be known rather than estimated, but it also means the specific magnitude of confounding and effect size tested here is one plausible scenario among many, not a re-derivation of any single real-world confounding structure.

4.2.2 The doubly-robust estimator is unstable in this sample size. The doubly-robust model fits seven parameters (treatment plus six covariates) on roughly 140–180 observed events; as the 500-seed Monte Carlo replication in Section 5.2 shows, this produces outright non-convergence in 11.6% of seeds and, among the seeds that do fit, an extreme, practically unusable hazard ratio in a further 3.6% of the original 500, both traced to small-sample multicollinearity. This instability is disclosed and used as an analytical finding in its own right, but it also means the doubly-robust estimate should be read as a secondary check, not as the pipeline's primary causal estimate, at this cohort size.

4.2.3 Misspecification analysis covers single-covariate omission only. The propensity-model misspecification sensitivity analysis (Section 3.5) sweeps all six confounders, one at a time, so the finding that omission severity depends on the presence of a correlated proxy is drawn from a complete rather than a partial set of examples. It still tests exactly one failure mode, however: omitting a single true confounder entirely from an otherwise correctly specified linear-logistic propensity model. It does not test wrong functional form (e.g. an omitted non-linearity or interaction among included covariates), omission of more than one confounder simultaneously, or measurement error in the covariates that are included. Real-world propensity models can fail in all of these additional ways [6], and this pipeline's degradation results should not be read as a bound on how badly IPTW can perform under an arbitrary form of misspecification.

4.2.4 Demographic covariates are extracted but not modeled. Biological sex and race are extracted alongside the five clinical covariates while building the complete-case cohort (Section 3), but neither is part of the injected confounding structure or either causal model. This is a deliberate scope choice, made so that the ground truth, the correlation analysis, and the misspecification sweep all refer to one fixed, fully enumerated six-covariate confounding structure rather than an open-ended one, but it does mean the pipeline does not speak to how these estimators would behave in the presence of demographic confounding, which is frequently material in real RWE studies of cardiometabolic drugs [13].

4.2.5 No competing-risks or mortality modeling. The outcome model treats heart failure hospitalization as the only event of interest, with all patients otherwise administratively censored at two years. Death from another cause before a heart failure hospitalization is not modeled as a competing risk; in a real elderly T2DM population, all-cause mortality is non-negligible over a two-year horizon and would compete with, rather than simply censor, the heart failure outcome [4]. A more complete model would separate these two fates rather than folding both into a single-event survival framework.

4.3 Results limitation

4.3.1 What the results are, restated precisely. The hazard ratios reported in Section 5 are estimates of an *investigator-injected*, simulated treatment effect, recovered from a semi-synthetic dataset whose covariates are Synthea-generated but whose treatment assignment and outcome are not. They are not:

- a new epidemiological finding about SGLT2 inhibitors and heart failure risk
- evidence that should influence real prescribing decisions

- a claim that Synthea's synthetic population reflects the true prevalence of type 2 diabetes, heart failure, or any comorbidity in a real population.

The correct reading of this project's results is methodological: under a known, disclosed confounding structure and a known true effect, this specific implementation of stabilized IPTW and doubly-robust estimation does, or does not, recover that effect, and does so with what empirical coverage rate. This is a statement about estimator behavior, not about SGLT2 inhibitors.

4.3.2 Generalization to real observational data is not demonstrated. Recovering a known effect from semi-synthetic data under a correctly-modeled (if partially misspecified, Section 3.5) confounding structure is a necessary, but not sufficient, demonstration that the same pipeline would perform equally well on real claims or EHR data. Several real-world complications are absent from this simulation entirely: unmeasured confounding beyond the six modeled covariates (the E-value analysis, Section 3.3, quantifies robustness to this within the simulated setting, but does not certify it for a real dataset with a different confounding structure), immortal time bias from cohort-entry and exposure-definition choices, informative censoring linked to disease severity, and outcome misclassification in diagnosis or procedure coding. Each of these is a well-documented threat to validity in real RWE studies [20] that this pipeline, by construction, does not face, and none of them should be assumed away when the same pipeline is eventually pointed at real data.

4.3.3 Doubly-robust instability is itself the argument for more modern alternatives. The instability documented above is the classical, textbook motivation for cross-fitted or machine-learning-based doubly-robust estimators (e.g. targeted maximum likelihood estimation or augmented IPTW with sample-splitting), designed specifically to avoid the overfitting and multicollinearity a single plug-in regression is prone to at this sample size [1]. This pipeline does not implement either alternative; doing so is noted as a natural extension in Section 6, not as a correction to the finding reported here.

5 RESULTS

This section presents the results of the pipeline, organized to answer the three nested questions framed in Section 1, at increasing levels of scrutiny:

(1) Recovery: does each estimator recover the injected hazard ratio in a single run, and how large is the confounding bias each corrects for?

(2) Calibration: across the 500-seed Monte Carlo replication, is each estimator's confidence interval well calibrated, and what failure modes only become visible under replication?

(3) Robustness: do the auxiliary diagnostics (negative control, E-value) and the misspecification sensitivity analysis support the primary result?

5.1 Recovery: single-run hazard ratio estimates

Table 6 reports the hazard ratio for SGLT2 inhibitor vs. DPP-4 inhibitor use on incident heart failure hospitalization, for each of the three estimators, against the known injected true value of 0.67.

Table 6: Primary run estimated hazard ratios

Model	HR	95% CI	Recovers truth?
Naive (unadjusted)	0.481	0.349 − 0.663	No
IPTW-weighted	0.496	0.351 − 0.702	Yes
Doubly-robust	0.482	0.329 − 0.706	Yes

The naive, unadjusted comparison substantially overstates the treatment benefit: its point estimate (0.481) is well below the true 0.67, and its confidence interval (0.349–0.663) excludes the true value entirely, confirming that the injected covariate-driven confounding is strong enough to produce a clearly misleading result if left uncorrected. Both IPTW and the doubly-robust estimator shift the point estimate back toward the true value and produce confidence intervals that contain it. IPTW also achieved full covariate balance after weighting, with all six covariates' standardized mean differences below the conventional $|SMD| < 0.1$ threshold, down from a pre-weighting imbalance as large as $|SMD| = 0.56$ on the most confounded covariate, visualized in the Love plot (Fig. 1).

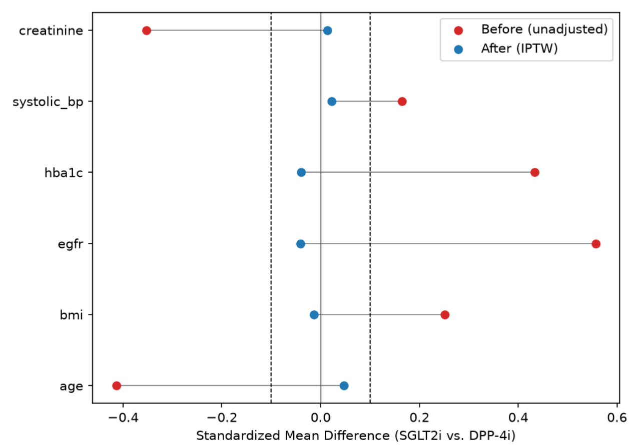

Figure 1: Standardized mean difference per covariate, before and after stabilized IPTW weighting, primary run. The dashed line marks the conventional $|SMD| = 0.1$ balance threshold.

5.2 Calibration: Monte Carlo validation

A single favorable run is not, by itself, evidence of a calibrated estimator. Table 7 summarizes each estimator's behavior across the 500-seed Monte Carlo replication described in Section 3.4.

Table 7: 500-seed Monte Carlo estimated hazard ratios

Model	Mean HR	Std. dev.	CI coverage
Naive (unadjusted)	0.523	0.088	70.6%
IPTW-weighted	0.671	0.155	94.6%
Doubly-robust	2.285	19.000	90.7%

IPTW's mean hazard ratio across the 500 seeds (0.671) is close to the true value of 0.670, and its empirical coverage rate (94.6%) sits close to the nominal 95% target, confirming that the single-run result of Section 5.1 was not a fluke: IPTW is a well-calibrated estimator under this data-generating process. The naive estimator's coverage of only 70.6% (against a 95% nominal target) confirms that its single-run bias is systematic, not a one-off unlucky draw.

The doubly-robust estimator's story, visible only under replication, involves two distinct failure modes. First, 58 of 500 seeds (11.6%) fail outright: the Cox fit raises a hard convergence error (e.g. a singular information matrix), and these seeds are dropped from Table 7 entirely. Second, among the 442 seeds that do fit, 18 more (3.6% of the original 500) converge to an extreme, practically unusable hazard ratio outside the [0.05, 5.0] bound: not a failed fit, but a misleading one returned quietly. Both trace to the same cause, small-sample multicollinearity from fitting seven parameters (treatment plus six covariates) on roughly 140–180 observed events, though a given seed can only exhibit one mode, not both. At this scale of replication the pathological-fit rate is a small minority of seeds, but it is not zero, and a handful of truly extreme outliers among them are large enough to dominate the doubly-robust estimator's mean (2.285, more than three times the true 0.670) and standard deviation (19.000, two orders of magnitude above IPTW's), since a simple mean is pulled far more by a handful of extreme values than a coverage rate computed seed-by-seed is.

Seed 1028 illustrates the pathological (not the failed) failure mode concretely: it fits successfully but produces a doubly-robust hazard ratio of 10.26 (95% CI 1.90–55.40), driven by a minimum estimated propensity score of 0.023 and a resulting maximum IPTW weight of 6.53, i.e. a single patient with a near-zero probability of their observed treatment dominating the weighted regression. This is a comparatively mild case: at the extreme end, seed 1123 fits but returns a doubly-robust hazard ratio of 377 with a 95% CI spanning roughly 10^{-10} to 10^{15}, a numerically self-evident non-answer rather than a merely large one. Accordingly, even setting aside the 11.6% of seeds dropped outright, the doubly-robust estimator's 90.7% empirical coverage (Table 7) still falls short of the 95% nominal target: its successfully-fitted seeds are not reliably well-calibrated either. Fig. 2 shows this directly: the IPTW distribution is tightly centered on the true hazard ratio, while the doubly-robust distribution is visibly right-skewed by a small number of extreme, pathological seeds like 1028 and 1123. On the strength of this replication evidence, the doubly-robust estimator is retained in this paper only as a disclosed secondary check, and IPTW is treated as the pipeline's primary causal estimator.
Fig. 3 shows the mean Love plot, covariate balance averaged across all 500 seeds rather than a single run, confirming that IPTW's balance property from Section 5.1 holds systematically and is not an artifact of a favorably-drawn primary seed.

5.3 Robustness: diagnostics and checks

Three further checks support the credibility of the IPTW result. First, a negative control outcome, sharing the same confounding structure but with a true hazard ratio fixed at 1.0, was re-analyzed

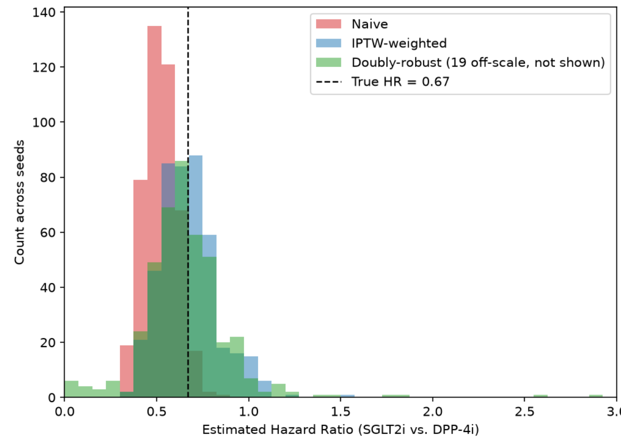

Figure 2: Distribution of estimated hazard ratios across the 500-seed Monte Carlo replication, by estimator.

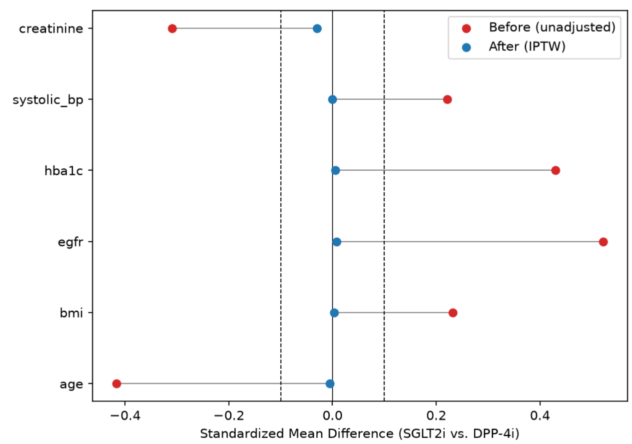

Figure 3: Standardized mean difference per covariate, before and after stabilized IPTW weighting, averaged across all 500 Monte Carlo seeds. The dashed line marks the conventional |SMD| = 0.1 balance threshold.

with both the naive and IPTW estimators: the naive estimate (HR 0.803, 95% CI 0.581–1.109) and the IPTW estimate (HR 1.112, 95% CI 0.723–1.711) both correctly contain the null, i.e. neither method manufactures a spurious effect where none was injected.

Second, the VanderWeele E-value computed from the IPTW-weighted result is 2.20: an unmeasured confounder would need at least this strength of association with both treatment and outcome, above and beyond the six measured covariates, to fully explain away the observed association at the confidence-limit bound closest to the null.

Third, Table 8 reports the full leave-one-out misspecification sensitivity analysis (Section 3.5): IPTW's mean hazard ratio and empirical coverage across the same 500-seed Monte Carlo procedure, but with each of the six true confounders, in turn, omitted from the estimated propensity model, ranked by absolute bias against the true hazard ratio of 0.67.

Table 8: IPTW under propensity-model misspecification

Propensity model	IPTW mean HR	Coverage	|Bias|
Full model	0.671	94.6%	0.001
Omitting age	0.597	92.4%	0.073
Omitting creatinine	0.630	93.1%	0.040
Omitting BMI	0.703	96.1%	0.033
Omitting eGFR	0.700	91.4%	0.030
Omitting systolic BP	0.683	93.7%	0.013
Omitting HbA1c	0.681	94.5%	0.011

Omitting any single confounder biases the mean IPTW estimate away from the true hazard ratio, but the six covariates are far from equally damaging. Age is clearly the worst covariate to omit, more than 1.8 times as damaging as the next-worst omission, consistent with Table 5: age carries the second-largest assignment coefficient in magnitude and has no strongly correlated proxy among the remaining five covariates (its best correlate is HbA1c at $r = -0.352$, Table 4), so its removal leaves a large share of the true confounding uncorrected with nothing left behind to compensate for it.

eGFR and creatinine carry the largest and fourth-largest assignment coefficients respectively (0.40 and −0.25, Table 5), so by coefficient magnitude alone eGFR should be the more damaging of the two to omit; instead the two land close together in a moderately-damaging middle tier (0.040 and 0.030), with creatinine's omission slightly the worse of the pair in this replication. Both are correlated with each other at $r = -0.546$ (Table 4), since eGFR is itself computed from serum creatinine via the MDRD formula rather than measured independently [8]: when either is dropped from the estimated propensity model, the one still included partially proxies for it, blunting (though not eliminating) the bias. The exact ordering between eGFR and creatinine is not itself a robust finding, since which of the two ranks worse depends on the covariate values re-extracted after the duplicate-observation fix (Section 3) and could plausibly reshuffle under a different resolution of that same tie; what is robust across both the pre- and post-fix data is that the pair sits together, well below age and well above HbA1c and systolic blood pressure.

Omitting HbA1c or systolic blood pressure is, by contrast, comparatively low-bias (0.011 and 0.013, the two least damaging omissions), which is more directly explained by their own assignment coefficients being the smallest of the six (0.05 and 0.10, Table 5) than by a correlated proxy standing in for them: systolic blood pressure's strongest correlate is a modest $r = -0.214$ with age, and its own estimated propensity coefficient is statistically indistinguishable from zero in the primary run ($p = 0.579$, Section 5.1), so there is comparatively little true confounding signal for its omission to remove in the first place.

The practical takeaway is accordingly two-fold rather than a single rule: a covariate is safe to omit either because a measured, correlated variable already stands in for it (eGFR, creatinine), or because it was never carrying much of the confounding to begin with (HbA1c, systolic blood pressure); a covariate with neither property, such as age here, is the one an analyst should be most reluctant to drop. Coverage tells a related but not identical story: every scenario shows coverage below the correctly specified baseline except omitting BMI, which combines the third-largest point-estimate bias (0.033) with a coverage *gain* (96.1%), its bias absorbed by a wider confidence interval rather than by a shift large enough to exclude the truth, a reminder that bias and coverage are related but not interchangeable diagnostics. Under the same sweep, the doubly-robust estimator's mean hazard ratio is extremely noisy across all six scenarios (1.26–2.67, with double-digit counts of pathological seeds in every scenario), far more unstable than under the correctly specified model documented in Section 5.2, and is reported for completeness rather than used to draw misspecification conclusions.

6 CONCLUSION

This project set out to answer a question that RWE methodology can rarely answer about itself: when a propensity-score causal pipeline reports a hazard ratio from confounded observational data, is that number actually close to the truth? Starting from real Synthea baseline covariates for 1,551 type 2 diabetes patients on metformin, the pipeline injects a known treatment assignment and outcome, with a true SGLT2i hazard ratio of 0.67 anchored to published cardiovascular-outcomes trials, and then runs the same naive, IPTW, and doubly-robust estimators an RWE analyst would run on real data, with the difference that here the right answer is known in advance.

The results behave the way a methodologically literate reviewer would expect. The naive comparison is clearly and substantially biased, exactly as strong covariate-driven confounding by indication should produce. Stabilized IPTW corrects that bias, achieves full covariate balance, and, critically, does so not just in one favorable run but across a 500-seed Monte Carlo replication with 94.6% empirical confidence-interval coverage against a 95% nominal target. The negative control and E-value analyses add convergent, if not conclusive, support. And the full leave-one-out misspecification sweep goes further than confirming IPTW degrades when a confounder is dropped: degradation severity tracks both a covariate's own effect size and whether a correlated proxy for it remains in the model (Section 3.5). Age, with a large effect and no proxy, is by far the worst to drop; creatinine, BMI, and eGFR form a moderately-damaging middle tier, with eGFR and creatinine partly buffering each other and BMI proving more damaging than eGFR precisely because it lacks a comparably strong proxy despite a smaller raw coefficient; and HbA1c and systolic blood pressure are safest to drop simply because they were never carrying much confounding to begin with.

The most valuable finding, however, is arguably the one that a single-run analysis would have missed entirely: the doubly-robust estimator, despite landing closest to the truth in the primary run, is shown by Monte Carlo replication to be unstable in this small-sample setting: it fails to fit outright in 11.6% of seeds, and even among the seeds that do fit, a further 3.6% of the original 500 converge to an extreme, practically unusable hazard ratio, both traced to small-sample multicollinearity rather than any more exotic cause. That finding would be invisible to an analyst who ran the pipeline once, got a plausible-looking doubly-robust number, and moved on, which is precisely the situation most real RWE analyses are in, since they cannot replicate against 500 independent draws of an unknown ground truth. Surfacing this instability here, where it can be directly attributed to a known cause rather than treated as unexplained variance, is the clearest demonstration of what a ground-truth-validated pipeline can offer that a single real-data analysis structurally cannot.

This validation exercise has one honest boundary, restated from Section 4: it demonstrates that this pipeline recovers a known effect under a disclosed, simulated confounding structure, not that the same pipeline would perform identically on real claims or EHR data, where unmeasured confounding, immortal time bias, informative censoring, and outcome misclassification are all live threats this simulation does not include. The natural next step is exactly the one this project's own instability finding points toward: replacing the plug-in doubly-robust estimator with a cross-fitted, machine-learning-based alternative such as targeted maximum likelihood estimation or augmented IPTW with sample-splitting, and testing whether that modern approach closes the stability gap this paper has documented, ideally under the same Monte Carlo discipline that made the gap visible in the first place.

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